

Testing VPC Connectivity

 alexten3517@gmail.com

[Option+S]

N. Virginia ▾

newproject10 @ 3904-0253-4868 ▾

VPC

IAM

EC2

CloudWatch

Amazon Lex

S3

```
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    selectedTrack=""
  >/project-app>

</body>
</html>
ec2-user@ip-10-0-0-147 ~$
```



Introducing Today's Project!

What is Amazon VPC?

Amazon Virtual Private Cloud (VPC) is a virtual network environment within the AWS cloud that you can use to launch AWS resources like EC2 instances, databases, and other services. It's essentially a private, isolated network that you can customize.

How I used Amazon VPC in this project

I was able to utilize my Amazon VPC in today's project to test connectivity between private and public servers. It allowed me to showcase my abilities in troubleshooting connectivity issues by editing security groups, ACL's and subnets.

One thing I didn't expect in this project was...

I did not expect to see how possible it is to allow and block the different types of traffic that is allowed to reach a VPC.

This project took me...

I spent a total of 1 hour on this project.



Connecting to an EC2 Instance

Connectivity means getting resources in our network to communicate with each other, and how well they can communicate delivered data to each other. Without connectivity resources in our network would not be able to communicate.

My first connectivity test was whether I could connect to my networks public EC2 instance.

```
aws
Click to go back, hold to see history
VPC IAM EC2 CloudWatch Amazon Lex S3
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Fri Nov 29 22:53:33 2024 from 18.206.107.29
ec2-user@ip-10-0-0-147 ~]$ ping 10.0.1.37
PING 10.0.1.37 (10.0.1.37) 56(84) bytes of data.
```

i-04803d152076cf9d7 (Nextwork public Server)

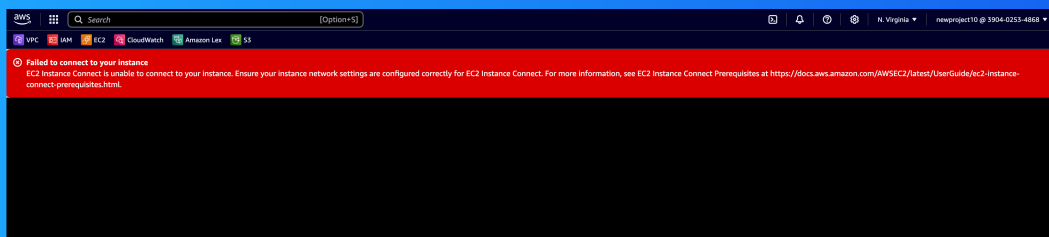


EC2 Instance Connect

I connected to my EC2 instance using EC2 Instance Connect, which is when one EC2 Instance makes a connection with another EC2 Instance without any errors. This allow for communication to happen between the two in a private way.

My first attempt at getting direct access to my public server resulted in an error, because my private server has a security group that did not allow SSH traffic, it only allow HTTP traffic a different protocol.

I fixed this error by adding a new inbound rule in my private servers security group that allowed SSH traffic from anywhere.





Connectivity Between Servers

Ping is a tool i used to test connectivity between two servers and also the response times. I used ping to test the connectivity between my public and private servers.

The ping command I ran was ping followed by the private ipv4 address for our private server which was 10.0.0.147

The first ping returned no replies from the private server. This meant security settings with my private server was blocking our inbound and outbound icmp traffic. Which is the traffic type of ping messages.

```
ec2-user@ip-10-0-0-147 ~$
```



Troubleshooting Connectivity

I troubleshooted this by allowing icmp traffic to flow in my private servers network ACL's and security groups.

```
aws
VPC IAM EC2 CloudWatch Amazon Lex S3
N. Virginia newproject10 @ 3904-0253-4868
last login: Sun Dec 1 01:28:21 2024 from 18.206.107.27
ec2-user@ip-10-0-0-147 ~]$ ping 10.0.1.37
PING 10.0.1.37 (10.0.1.37) 56(84) bytes of data.
 4 bytes from 10.0.1.37: icmp_seq=638 ttl=127 time=0.340 ms
 4 bytes from 10.0.1.37: icmp_seq=639 ttl=127 time=0.783 ms
 4 bytes from 10.0.1.37: icmp_seq=640 ttl=127 time=1.13 ms
 4 bytes from 10.0.1.37: icmp_seq=641 ttl=127 time=0.847 ms
 4 bytes from 10.0.1.37: icmp_seq=642 ttl=127 time=1.35 ms
 4 bytes from 10.0.1.37: icmp_seq=643 ttl=127 time=0.933 ms
 4 bytes from 10.0.1.37: icmp_seq=644 ttl=127 time=1.16 ms
 4 bytes from 10.0.1.37: icmp_seq=645 ttl=127 time=0.454 ms
 4 bytes from 10.0.1.37: icmp_seq=646 ttl=127 time=1.16 ms
 4 bytes from 10.0.1.37: icmp_seq=647 ttl=127 time=0.373 ms
 4 bytes from 10.0.1.37: icmp_seq=648 ttl=127 time=0.701 ms
 4 bytes from 10.0.1.37: icmp_seq=649 ttl=127 time=0.696 ms
 4 bytes from 10.0.1.37: icmp_seq=650 ttl=127 time=0.462 ms
 4 bytes from 10.0.1.37: icmp_seq=651 ttl=127 time=0.662 ms
 4 bytes from 10.0.1.37: icmp_seq=652 ttl=127 time=1.52 ms
 4 bytes from 10.0.1.37: icmp_seq=653 ttl=127 time=0.719 ms
 4 bytes from 10.0.1.37: icmp_seq=654 ttl=127 time=0.729 ms
 4 bytes from 10.0.1.37: icmp_seq=655 ttl=127 time=1.12 ms
 4 bytes from 10.0.1.37: icmp_seq=656 ttl=127 time=0.691 ms
 4 bytes from 10.0.1.37: icmp_seq=657 ttl=127 time=0.574 ms
 4 bytes from 10.0.1.37: icmp_seq=658 ttl=127 time=0.883 ms
 4 bytes from 10.0.1.37: icmp_seq=659 ttl=127 time=1.52 ms
 4 bytes from 10.0.1.37: icmp_seq=660 ttl=127 time=0.975 ms
 4 bytes from 10.0.1.37: icmp_seq=661 ttl=127 time=0.390 ms
 4 bytes from 10.0.1.37: icmp_seq=662 ttl=127 time=1.63 ms
 4 bytes from 10.0.1.37: icmp_seq=663 ttl=127 time=0.447 ms
 4 bytes from 10.0.1.37: icmp_seq=664 ttl=127 time=0.592 ms
 4 bytes from 10.0.1.37: icmp_seq=665 ttl=127 time=0.905 ms
 4 bytes from 10.0.1.37: icmp_seq=666 ttl=127 time=0.875 ms
C
--- 10.0.1.37 ping statistics ---
66 packets transmitted, 29 received, 95.6456% packet loss, time 691503ms
```



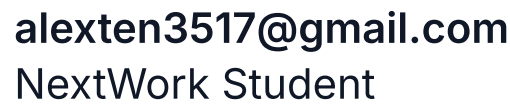
Connectivity to the Internet

Curl is a command we use to see if we can communicate with internet. The same we use ping to communicate with another server curl allow us to see if there is a response from a internet host.

I used curl to test if the connectivity of my public server, security group and network ACL was set up properly. Access to the webpage showed there is communication with the internet.

Ping vs Curl

Ping allows test network connectivity and measure latency (round-trip time) between two devices. While curl is primarily used to fetch data from web servers, test APIs, and simulate HTTP requests.



Connectivity to the Internet

aws

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Search

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ec2-user@ip-10-0-0-147 ~\$



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